

EduGenie: Google Gemini Powered Learning Assistant

6. Project Testing

This document defines the functional testing scope explicitly documented for EduGenie and organizes the source's testing activities into a structured validation checklist.

6.1 Testing Objective

The primary testing objective is to confirm that each documented learner-facing feature can be reached through the web interface and connected FastAPI endpoint.

6.2 Environment Test

Verify Python 3.10+ is installed using `python --version`.

Verify required framework and server dependencies are installed.

Verify the Gemini API key is configured for Gemini-powered functions.

Start the application using `uvicorn main:app --reload`.

6.3 Application Access Test

Open `http://127.0.0.1:8000` in a browser.

Confirm the EduGenie interface loads.

Confirm the task dropdown, text area, submit button, and result area are available.

6.4 Q&A; Test

Enter an educational question.

Select QnA.

Submit the request.

Verify that an answer is returned through the `/qa` endpoint.

6.5 Explanation Test

Enter a concept such as a documented academic topic.

Select Explain.

Submit the request.

Verify that a simplified explanation is displayed through `/explain`.

6.6 Quiz Test

Provide a topic or passage.

Select Quiz.

Verify that three MCQs are generated.

Verify that each question contains four options and a correct answer.

If parsing fails, verify that the system returns a useful error rather than silently failing.

6.7 Summary Test

Provide a long educational paragraph.

Select Summary.

Verify that the response is concise while retaining core information.

6.8 Learning Recommendation Test

Enter a topic such as SQL.

Select Recommend Path.

Verify that the output is organized from beginner to advanced and includes learning guidance/resources as documented.

6.9 Frontend Integration Test

Verify that form submission sends a POST request to the backend.

Verify that the result appears below the input without requiring a separate page.

6.10 Error-Handling Focus

Quiz generation and parsing errors are explicitly documented as handled with detailed debugging messages.

The learning-path module is documented to handle invalid or incomplete API responses.

Conclusion

Testing is feature-based and directly traces to the documented functional testing activities: question answering, explanation, quiz generation, summarization, and personalized learning recommendations.

Source basis: EduGenie project document supplied with this request and the repository structure shown in the supplied reference image. Where the source did not specify an implementation detail, this documentation preserves the documented scope rather than inventing project facts.